GLA UNIVERSITY

Department of Computer Engineering & Applications

MINI PROJECT – 1 SYNOPSIS

1. BASIC PROJECT DETAILS:

Project Title	Wellness Challenges & Leaderboard Experience
Batch No.	10
Section	3SA
Semester	V
Date of Submission	18 September 2026
Faculty Mentor / Guide	Satyam Kumar Jha (Integraminds)
Github Repository Link	https://github.com/viveksinghisyour-max/wellness_challenges_leaderboard

2. TEAM MEMBERS & ROLES:

S.No.	Name of Student	University Roll No.	Role in Project
1	Utkarsh Sharma	2415001708	Frontend Development & UI/UX
2	Varun Sharma	2415001756	Dashboard, Data Visualization & Animations
3	Vivek Kashyap	2415001808	Dataset Generation, Testing & Documentation
4	Vivek Kumar	2415001813	Team Lead, Core Logic & Integration

3. PROJECT DESCRIPTION:

Wellness Challenges & Leaderboard Experience is a community-wellness web application designed to encourage healthy habits through challenge-based participation, progress visualization, and friendly competition. Users can explore wellness challenges, view goals and duration, enroll in selected challenges, and monitor their performance through an interactive dashboard. The application can support activities such as steps, active minutes, hydration, mindfulness, and consistency-based goals. Animated progress rings, charts, badges, streaks, ranks, and achievement indicators will make activity information easier to understand and more engaging.

A key technical feature is a client-side leaderboard generated from a bundled or generated activity dataset. The dataset will contain participant records, challenge identifiers, dates, and activity measures. JavaScript logic will filter records, calculate challenge scores, sort participants, assign ranks, and update the leaderboard in the browser. The project also includes a simple anti-cheat-style visual flagging mechanism. Local rule checks will identify statistically implausible records, such as extremely high activity values or unusual combinations of steps and active time. Such records will be marked as unusual or requiring review rather than automatically being declared fraudulent. The project therefore combines frontend development, data processing, visualization, gamification, and transparent rule-based anomaly detection.

4. OBJECTIVES & PROPOSED WORK (WHAT WE ARE GOING TO DO):

The proposed work will be implemented as a modular client-side application. The team will first design the user interface and reusable components, then develop challenge enrollment, activity processing, visualization, leaderboard calculation, anomaly flagging, testing, and deployment. The objective is to produce a clear, interactive, and technically demonstrable prototype rather than only a static fitness webpage.

4.1 Functional Objectives:

1. **Challenge Discovery and Enrollment:** Provide a catalogue of wellness challenges with challenge name, description, goal, duration, participant count, status, and Join/Enroll functionality.
2. **Personal Dashboard:** Show active challenges, progress, activity summaries, current rank, achievements, streaks, and important wellness indicators.
3. **Progress Visualization:** Use animated circular progress rings and charts to represent progress toward steps, active minutes, hydration, mindfulness, and other measurable goals.
4. **Activity Dataset:** Prepare a bundled/generated JSON dataset containing participant IDs, names, challenge IDs, dates, steps, active minutes, hydration, calories, and related activity values.
5. **Client-Side Leaderboard:** Filter activity by challenge, calculate scores, sort participants, assign ranks, handle ties, and render the results dynamically inside the browser.
6. **Anomaly Flagging:** Apply transparent local rules to identify extreme values, unusual step-to-time relationships, and large activity changes, then show an understandable warning.
7. **Gamification:** Use XP, badges, streaks, ranks, achievement messages, and challenge-completion feedback to improve engagement.
8. **Responsive Interface:** Make the application usable on desktop and mobile screen sizes with consistent navigation, readable statistics, and clear status indicators.

4.2 Proposed Challenge Categories:

- 7-Day Step Challenge – a weekly target based on cumulative steps.
- Active Minutes Challenge – a target for total active minutes during the challenge period.
- Hydration Challenge – a daily or weekly hydration goal represented through progress tracking.
- Mindfulness Challenge – a target based on completed mindfulness or meditation minutes.
- Consistency Challenge – rewards regular participation and completion of daily wellness targets.

4.3 Current Stage of Work:

Current stage: project planning and interface/system design. The immediate implementation work will include project setup, component creation, dataset preparation, leaderboard and scoring logic, anomaly rules, testing with normal and unusual records, and deployment of the working prototype.

4.4 PROPOSED METHODOLOGY AND SYSTEM WORKFLOW:

The system will follow a component-based architecture. React with Vite will provide the application structure, Tailwind CSS will be used for responsive styling, and JavaScript/TypeScript will handle client-side processing. Activity data will be loaded from a bundled JSON dataset or generated locally. Utility functions will perform enrollment management, progress calculations, score calculation, leaderboard ranking, and anomaly checks. Processed results will be displayed through reusable dashboard, chart, progress-ring, challenge-card, and leaderboard components.

4.5 Planned Data Flow:

1. Step: Load the bundled/generated activity dataset when the application starts.

2. Step: Identify the selected challenge and relevant participant records.

3. Step: Calculate each participant's progress toward the challenge goal.

4. Step: Calculate the challenge score using a defined scoring rule.

5. Step: Sort the participants and assign leaderboard ranks.

6. Step: Run local anomaly rules against activity records.

7. Step: Attach a warning status and reason to records that meet anomaly conditions.

8. Step: Render progress rings, charts, leaderboard rows, badges, and activity summaries.

9. Step: Allow the user to inspect unusual entries without preventing normal participation.

4.6 Proposed Module Structure:

User Interface Module: Landing page, navigation, reusable cards, dashboard layout, responsive styling, and accessibility considerations.

Challenge Module: Challenge catalogue, challenge details, goals, duration, participation status, and enrollment handling.

Activity Module: Daily activity records, summaries, progress calculations, and trend visualization.

Leaderboard Module: Challenge filtering, score calculation, sorting, ranking, tie handling, and participant status.

Anomaly Module: Threshold checks, relationship checks, warning levels, and human-readable review reasons.

Gamification Module: XP, badges, streaks, rank indicators, achievement messages, and completion feedback.

Storage Module: localStorage for prototype enrollment/preferences and JSON for the bundled/generated activity dataset.

4.7 Scoring and Leaderboard Logic:

For a step-based challenge, the primary score may be the participant's total steps. For a multi-metric wellness challenge, a normalized weighted score can be used. A possible prototype configuration is steps 40%, active minutes 30%, hydration 20%, and consistency 10%. The exact weights can be adjusted during implementation. After scores are calculated, participants will be sorted in descending order and assigned ranks. Equal scores can receive the same rank or be resolved using a documented secondary metric.

4.8 Local Anomaly / Anti-Cheat-Style Checks:

The project will use simple, transparent rules rather than claiming to provide a complete fraud-detection system. Example checks include: unusually high daily step counts; unusually high active minutes; very high steps combined with very low active time; and extreme changes compared with a participant's previous activity. Multiple triggered rules can be combined into a warning level. The interface will use wording such as 'Unusual Activity' or 'Requires Review' so that a statistical flag is not presented as proof of cheating.

5. FUTURE SCOPE / ENHANCEMENTS:

The initial prototype can be expanded into a full community wellness platform. Future work can improve data collection, personalization, security, community interaction, analytics, and anomaly detection while retaining the transparent challenge and leaderboard experience.

1. **Wearable and Fitness API Integration:** Import activity automatically from supported fitness and wearable platforms.
2. **Authentication and Cloud Storage:** Add secure accounts, user profiles, synchronized challenge participation, and persistent cloud data.
3. **Advanced Anomaly Detection:** Complement local rules with statistical techniques and configurable review workflows.
4. **Social and Team Challenges:** Allow users to create teams, invite friends, share goals, and participate in group challenges.
5. **Personalized Challenge Recommendations:** Recommend challenges based on previous activity, completed challenges, and user-defined goals.
6. **Notifications and Reminders:** Provide reminders for daily targets, challenge deadlines, streak continuation, and achievements.
7. **Mobile/PWA Support:** Provide an installable and responsive mobile experience with appropriate offline support.
8. **Admin and Analytics Dashboard:** Provide challenge management, participant analytics, activity summaries, and tools for reviewing unusual records.

6. TOOLS & TECHNOLOGIES USED:

Technology Area	Proposed Tool / Technology
Frontend	React.js
Programming	JavaScript
Styling	Bootstrap
Animation	SVG, Framer Motion
Data Visualization	Recharts or equivalent charting library
Data Format	JSON bundled/generated activity dataset
Client Storage	Browser localStorage for prototype state
Icons / UI	Lucide React or equivalent icon library
Version Control	Git and GitHub
Testing	Functional testing, browser developer tools, responsive testing, normal/unusual test datasets
Deployment	Vercel

7. EXPECTED OUTCOME:

The expected outcome is a functional, responsive, and visually engaging wellness challenge web application. Users will be able to browse available challenges, enroll in a selected challenge, view activity and progress, and compare performance with other participants through a dynamically generated leaderboard.

From a technical perspective, the project will demonstrate reusable React components, client-side dataset processing, challenge-specific scoring, ranking, animated visualization, local storage, and transparent rule-based anomaly flagging. The final demonstration will include both normal activity records and selected statistically unusual records so that the team can show how the warning mechanism operates.

7.1 Expected Deliverables:

- Responsive web application prototype.
- Bundled/generated sample wellness activity dataset.
- Challenge enrollment and progress-tracking interface.
- Client-side leaderboard and scoring engine.
- Rule-based anomaly flagging with visual status indicators.
- Testing results covering normal, boundary, and unusual activity records.
- Project source code, README/documentation, and deployed application.

7.2 Project Success Criteria:

- Users can view and enroll in available challenges.
- Progress values and animated rings update correctly from activity data.
- Leaderboard rankings are calculated consistently from the dataset.
- Anomaly rules flag configured unusual records and provide understandable reasons.
- Interface remains usable across common desktop and mobile screen sizes.

• Core functionality can be demonstrated without requiring a complex backend.

Signature of Faculty Mentor: _____